SECTION 1

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. PURIFICATION: (A) heterogeneity (B) negligence (C) contamination (D) experimentation (E) infiltration

2. ABOLISH: (A) stretch (B) soothe (C) refurbish (D) establish (E) straighten

3. RUPTURE: (A) prove (B) regret (C) delay (D) scale down (E) join together

4. DESOLATE: (A) typical (B) populous (C) enormous (D) full of water (E) likely to change

5. ENTHRALLING: (A) incipient (B) boring (C) incomplete (D) minute (E) rejuvenating

6. DESPONDENCY: (A) blithe hopefulness (B) careful attention (C) quick reaction (D) proud dignity (E) bitter jealousy

7. EXPEDITIOUS: (A) protracted (B) pretentious (C) transcendent (D) sensitive and shy (E) slim and graceful

8. PROPRIETY: (A) gentility (B) generosity (C) indecorum (D) servility (E) inability

9. PARSIMONY: (A) formality (B) inhumanity (C) obesity (D) arrogance (E) extravagance

10. RECALCITRANT: (A) repetitious (B) meandering (C) determined (D) indecisive (E) compliant

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. On the assumption that planets and meteors were ---- as a result of the same cosmic catastrophe, their ---- should be roughly similar.

(A) formed. .compositions
(B) discovered. .dates
(C) evolved. .histories
(D) released. .shapes
(E) destroyed. .configurations

12. Advertising alone, no matter how ----, cannot convince people to ---- an item that answers no real and vital need.

(A) stringent. .ignore (B) innocuous. .modify (C) outrageous. .disregard (D) extensive. .purchase (E) enigmatic. .want

13. Perhaps part of the ---- of adolescent behavior is traceable to the violent swing between being dependent and wanting to be independent.

(A) capriciousness (B) predictability (C) reticence (D) irony (E) maturity

14. Self-consciously and ----, but insistently, philosophers keep asking these traditional questions.

(A) confidently (B) diffidently (C) frequently (D) pugnaciously (E) progressively

15. The artistic success of ancient vase painting did not depend entirely on the artist's skill in ---- imitation of observed appearances.

(A) slavish (B) forceful (C) incompetent (D) potential (E) creative

GO ON TO THE NEXT PAGE

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Princeton, N.J. 08541

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. PRACTICE : IMPROVEMENT :: (A) memory : event
(B) chaos : arrangement (C) assignment : praise
(D) thought : regret (E) polish : sparkle

17. BREATHE : PANT :: (A) borrow : pay
(B) drop : bounce (C) stride : gallop
(D) spend : save (E) scare : leap

18. LEMON : FRUIT :: (A) author : symposium
(B) branch : tree (C) poetry : prose
(D) hammer : tool (E) partner : syndicate

19. SCRAMBLE : WALK :: (A) race : compete
(B) scrawl : write (C) speak : state
(D) rebel : disagree (E) attack : engage

20. GENE : HEREDITY :: (A) germ : disease
(B) blood : vein (C) cell : nourishment
(D) pain : nerve (E) strength : muscle

21. MERGER : CORPORATIONS ::
(A) budget : finances
(B) indenture : services
(C) marriage : persons
(D) transaction : dealers
(E) annihilation : countries

22. DEBATE : ARGUMENT :: (A) debut : debutante
(B) illness : convalescence (C) campaign : expedition
(D) duel : fight (E) audition : performance

23. INCANTATION : MAGICAL ::
(A) idiom : persuasive
(B) lisp : ungrammatical
(C) phrase : definitional
(D) endearment : affectionate
(E) nomenclature : random

24. MELODEON : ORGANIST ::
(A) reveille : bugler
(B) solo : accompanist
(C) crescendo : pianist
(D) anthem : choirmaster
(E) kettledrum : tympanist

25. LINGUISTICS : LANGUAGE ::
(A) statistics : sociology
(B) ceramics : clay
(C) gymnastics : health
(D) dynamics : motion
(E) economics : warfare

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Archaeology, by its very nature, moves at a slow and deliberate pace, yet it is in the midst of a quiet revolution today. No longer does the study of history confine itself to quests for exotic art objects that glorified rulers in times past. Today, archaeologists are trying to put together a working picture of how ancient societies—particularly the common people in those societies—behaved.

This fresh approach embraces a number of separate strands. Sophisticated instruments developed by physical scientists—computers, nuclear sensors, and even cameras aboard earth satellites—are helping archaeologists to identify and explore promising sites that will yield artifacts for them to interpret. Techniques gleaned from sociology and psychology are supplying archaeologists with reams of statistical data. Some modern archaeologists do not even bother to dig anymore. They prefer to study such relatively modern monuments as gravestones and latter-day examples of primitive peoples like the aborigines of Australia and the Bushmen of Botswana.

The old and the new archaeologists, however, share one fundamental assumption: that modern society can learn from the mistakes of its ancestors. By analyzing how ancient societies coped with such problems as pollution, overpopulation, scarcity of resources, and warfare, archaeologists believe that they can suggest means to help deal with the modern versions of these social scourges.

26. Which of the following would be the most appropriate title for the passage?

(A) Past and Present Archaeological Techniques
(B) The Objectives of Archaeology: Predicting the Future
(C) Critical Attack on the New Archaeology
(D) Finding New Sites for Archaeological Exploration
(E) A New Approach to Archaeology

27. The author provides information that would answer all of the following questions EXCEPT:

(A) Do archaeological studies have implications for scientists dealing with problems of modern-day man?
(B) What sort of studies would an archaeologist following the new approach undertake?
(C) What kind of devices are used to aid the new archaeology?
(D) What are some of the views held by archaeologists on the new archaeology?
(E) What are some of the goals of archaeology?

28. The author's attitude toward the new approach to archaeology would best be described as

(A) sentimental
(B) condescending
(C) approving
(D) pedantic
(E) apathetic

GO ON TO THE NEXT PAGE

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

Perhaps it is a kind of ethnic chauvinism that is responsible for the fact that many physiologists and nutritionists were surprised to discover that a large percentage of the human adult population is deficient in lactase, the enzyme that breaks down lactose (milk sugar). The few human populations in which lactose tolerance exceeds intolerance include most Northern European and white American ethnic groups.

A period of active investigation of lactose intolerance was initiated at the end of the 1950's. In 1965 investigators who had administered lactose to American blacks and whites reported some startling findings: whereas only fifteen per cent of the whites showed clinical symptoms of lactose intolerance, about seventy per cent of the blacks were lactose intolerant.

In the study various methods of measuring intolerance were used. One was recording the appearance of clinical symptoms after the ingestion of a dose of lactose, which was standardized at two grams of lactose per kilogram of body weight up to a maximum of 100 grams. Another measured low lactase activity, determined through an intestinal biopsy, after ingestion of the lactose. A third recorded the elevation of blood glucose after ingestion of the lactose.

Since clinical symptoms are variable, and the biopsy method is inconvenient for the subject being tested, the blood-glucose method was preferred. It is a direct measure of lactose breakdown, and false negative results are rare if the glucose is measured fifteen minutes after lactose is administered.

By 1970 enough data had been accumulated to indicate that more people all over the world are intolerant to lactose than are tolerant.

29. On the basis of the information in the passage, which of the following conclusions can be made about the participants in the 1965 study?

(A) The majority of the participants were black.
(B) The majority of the participants were white.
(C) None of the participants exhibited clinical symptoms of lactose intolerance.
(D) A majority of the blacks were intolerant to lactose.
(E) A majority of the whites had a lactase deficiency.

30. The main purpose of the passage is to

(A) report on the use of a new technical procedure
(B) challenge the significance of recent experimental data
(C) describe the investigation of a common condition
(D) analyze the reasons for a medical phenomenon
(E) describe the symptoms of a human ailment

31. According to the passage, a person receiving the maximum dosage of lactose would weigh at least

(A) 10 kilograms
(B) 50 kilograms
(C) 100 kilograms
(D) 150 kilograms
(E) 200 kilograms

32. The author suggests that prior to the late 1950's, scientists were unaware of the common occurrence of lactose intolerance in adults because

(A) lactose intolerance was not prevalent among white ethnic groups
(B) they did not know the symptoms of lactose intolerance
(C) they had not identified the function of lactase in human digestion
(D) they did not know the source of lactose in the human diet
(E) it was not yet possible to measure enzyme activities

GO ON TO THE NEXT PAGE

I call myself a conservationist and am therefore anxious to preserve the environment, particularly my personal environment, and most especially the maple trees in front of my house. One day, sitting in my study, devising strategies of how to deal with "them" when they came on their next, inevitable visit—I heard them. They were here! The snarling of the chain saw was a reality and not just a figment of my imagination.

I leapt from my chair and ran around the corner of the house to the group of men standing around the truck, looking up at the elevated basket that carried the workman with the chain saw. He was cutting branches off my maple trees as if they were so many bunches of bananas.

"What do you think you're doing?"

"Pruning your trees," said a man who looked as if he might be in charge.

"Why?"

He told me that they needed to put an additional set of wires along the poles. "What is it?" I was still shouting. "Another subdivision going up? More houses going to be built? Because if . . ."

"No," he interrupted evenly. "We just need to get more electricity to the houses already here." Defeated, I turned back to the old farmhouse with its electric toothbrush, self-cleaning oven, drier, and electric can opener.

I had met the enemy.

33. The passage suggests that the workmen's primary objective is to

(A) increase the supply of electricity in the narrator's neighborhood
(B) comply with the wishes of the narrator
(C) preserve the trees in the area
(D) provide a source of electricity for a planned housing development
(E) replace the narrator's trees

34. It can be inferred from the passage that the

(A) workmen did not want the narrator to know that they were pruning the trees
(B) man in charge of the work crew sympathized with the narrator
(C) workmen were antagonistic toward the narrator
(D) maple trees had been pruned without the narrator's permission on previous occasions
(E) workman cutting the branches off the trees was planning to cut the trees down

35. The enemy referred to in the last sentence is probably

(A) the workman using the chain saw
(B) the narrator
(C) the narrator's neighbors
(D) builders of new developments
(E) manufacturers of electrical equipment

GO ON TO THE NEXT PAGE

To emphasize the stagnation and the narrowness of the society depicted in Jane Austen's novels is to take a narrow and mechanical view of them. *Emma* is not a period piece, nor is it what is sometimes called a "comedy of manners." We read it to illuminate not only the past but also the present. And we must face here in both its crudity and its importance a question: Exactly what relevance and helpfulness does *Emma* have for us today? In what sense does a novel dealing skillfully and realistically with a society and its standards, which are dead and gone forever, have value in our very different world today? Stated in such terms, the question itself is unsatisfactory. If *Emma* today captures our imagination and engages our sympathies (as, in fact, it does), then either it has some genuine value for us, or else there is something wrong with the way we give our sympathy and our values are pretty useless.

Put this way, it is clear that anyone who enjoys *Emma* and then remarks "but of course it has no relevance today" is, in fact, debasing the novel, looking at it not as a living, enjoyable work of art but as a mere dead picture of a past society. Such an attitude is fatal both to art and to life. It can be assumed that *Emma* has relevance. The helpful approach is to ask why this novel still has the power to move us today.

What gives *Emma* its power to move us is the realism and depth of feeling behind Jane Austen's attitudes. She examines with a scrupulous yet passionate and critical precision the actual problems of her world. That this world is narrow cannot be denied. But the value of a work of art rests on the depth and truth of the experience it communicates, and such qualities cannot be identified with the breadth of the work's panorama. A conversation between two people in a grocery store may tell us more about a world war than a volume of dispatches from the front. The silliest of all criticisms of Jane Austen is the one that blames her for not writing about the Battle of Waterloo and the French Revolution, which were in the headlines of the newspapers she read. She wrote about what she genuinely understood, and no artist can do more.

36. The main idea of the passage is that

(A) a constricted view of *Emma* is natural and acceptable
(B) a novel should not depict a vanished society
(C) a good novel is an intellectual rather than an emotional experience
(D) many readers have seen only the comedy in *Emma*
(E) *Emma* should be read with sensitivity and an open mind

37. The author would probably disagree with those critics or readers who find that the society in Jane Austen's novels is

(A) unsympathetic (B) uninteresting (C) crude (D) authoritarian (E) provincial

38. The author implies that a work of art is properly judged on the basis of its

(A) universality of human experience truthfully recorded
(B) popularity and critical acclaim in its own age
(C) openness to varied interpretations, including seemingly contradictory ones
(D) avoidance of political and social issues of minor importance
(E) continued popularity through different eras and with different societies

39. It can be inferred that the author considers the question stated and restated in lines 8-13 to be unsatisfactory because it

(A) fails to assume that society and its standards are the proper concern of a novel
(B) neglects to assume that the novel is a definable art form
(C) suggests that our society and Jane Austen's are quite different
(D) fails to emphasize Jane Austen's influence on modern writers
(E) wrongly states the criteria for judging a novel's worth

40. The author's attitude toward someone who "enjoys *Emma* and then remarks 'but of course it has no relevance today'" (lines 21-22) can best be described as one of

(A) amusement
(B) astonishment
(C) disapproval
(D) resignation
(E) ambivalence

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 3

Time—30 minutes

25 QUESTIONS

In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

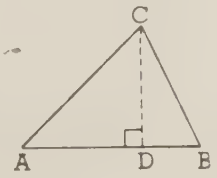
Circle of radius r: Area = πr^2; Circumference = $2\pi r$

The number of degrees of arc in a circle is 360.

The measure in degrees of a straight angle is 180.

Definitions of symbols:

$=$	is equal to	$\leqslant$	is less than or equal to
$\neq$	is unequal to	$\geqslant$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $x + y = 2$, then $x + y - 4 =$

   (A) -2 (B) 0 (C) 2 (D) 4 (E) 6

2. $14 \div 3\frac{1}{2}$ is

   (A) $\frac{1}{49}$

   (B) $\frac{1}{4}$

   (C) 5

   (D) 49

   (E) none of the above

3. According to the rules of an organization that has 100 members, no member may serve on more than 3 committees. What is the maximum permissible number of 10-man committees?

   (A) 30
   (B) 29
   (C) 21
   (D) 20
   (E) 10

4. If n is a positive integer greater than 1 and if the remainders are the same when 9 and 16 are each divided by n, then $n =$

   (A) 2
   (B) 5
   (C) 6
   (D) 7
   (E) 8

5. How many tenths of a meter are there in 21.1 meters?

   (A) 0.1 (B) 1 (C) 2.11 (D) 21.1 (E) 211

6. Which of the following is NOT equal to the ratio of two whole numbers?

   (A) $\frac{1}{3}$

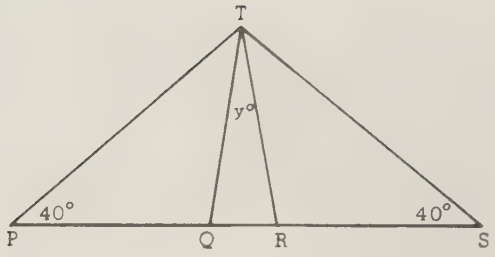
   (B) $\frac{3^2}{1}$

   (C) 0.3

   (D) 3%

   (E) $\frac{\sqrt{3}}{1}$

GO ON TO THE NEXT PAGE



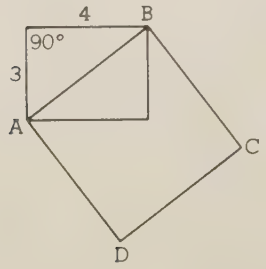
7. In the figure above, if PS is a line segment and PQ = QT = TR = RS, then y =

(A) 10 (B) 20 (C) 30 (D) 40 (E) 50

---

8. A shipment of 1,680 items is trucked across states M and N. If State M levies a tax of 7 cents for each lot of 7 items and if State N levies a tax of 5 cents for each lot of 5 items, what is the total tax levied by the two states?

(A) $5.76
(B) $33.60
(C) $35.52
(D) $74.00
(E) $576.00



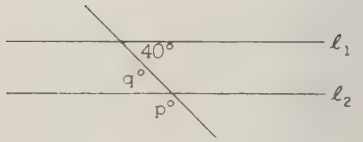
9. In the figure above, what is the length of a side of square ABCD ?

(A) 5 (B) 6 (C) 12 (D) 13 (E) 25

---

10. Colored banners are strung on a cord to advertise a carnival. If the colors form a repeating pattern starting with white, red, yellow, blue, green, purple; white, red, yellow, blue, green, purple, and so on, what is the color of the 76th banner?

(A) Red
(B) Yellow
(C) Blue
(D) Green
(E) Purple



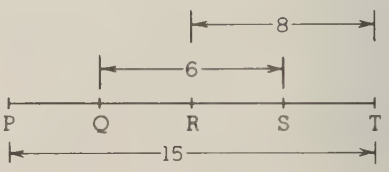
11. In the figure above, if $\ell_1 \parallel \ell_2$, then p − q =

(A) 40
(B) 70
(C) 100
(D) 110
(E) 140

---

12. If x and y are integers and if $x - y > x + y$, which of the following must be true?

(A) $y < 0$
(B) $x < 0$
(C) $x = y$
(D) $y > x$
(E) $x > y$



Note: Figure not drawn to scale.

13. In the figure above, if R is the midpoint of QS, then PQ =

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

---

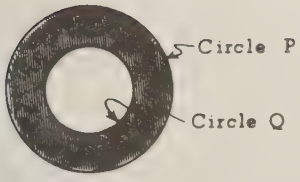
14. In a set of 24 cubic blocks, each block has pictures on two sides and letters on the remaining sides. How many sides of the entire set have letters?

(A) 24 (B) 48 (C) 72 (D) 96 (E) 144

15. If $p = 27 + t$, $r = 27 - t$, and $t^2 = 9$, then $\frac{pr}{9}$ =

(A) 2
(B) 26
(C) 72
(D) 80
(E) 81

GO ON TO THE NEXT PAGE



16. In the figure above, if circle P has radius R and circle Q has radius r, the area of the shaded region is

(A) $\pi(R^2 + r^2)$ (B) $\pi(R^2 - r^2)$ (C) $\pi(R - r)^2$
(D) $2\pi(R - r)$ (E) $2\pi(R + r)$

---

17. In a certain country the ratio of the number of people 30 years of age or under to the number of people over 30 is 3 to 2. What per cent of the population of this country is over 30 ?

(A) 40%

(B) 50%

(C) 60%

(D) $66\frac{2}{3}\%$

(E) $88\frac{1}{3}\%$

18. Three prizes, with an average value of $75.00, were offered in a poster contest. However, the judges chose 2 first-place winners and 2 second-place winners and decided to split the prize money so that each first prize was twice as much as each second prize. How much was awarded to each first-place winner?

(A) $25.00 (B) $37.50 (C) $62.50
(D) $75.00 (E) $100.00

19. Tom and Joe together earn $750 per week. If Tom's salary is two-thirds of Joe's, what is three-fourths of Tom's weekly salary?

(A) $187.50
(B) $225.00
(C) $275.00
(D) $337.50
(E) $375.00

20. If $\frac{k}{7} = \frac{m}{8}$, $k > 0$, and $m > 0$, which of the following must be true?

(A) $m < k$
(B) $k < m$
(C) $k = 7, m = 8$ only
(D) $k = 8, m = 7$ only
(E) None of the above

21. If a man reads for 5 hours and x minutes at an average rate of 40 printed pages an hour, then, in terms of x, he will read exactly how many printed pages?

(A) 200x

(B) 200 + 40x

(C) $200 + \frac{2}{3}x$

(D) 200 + 2400x

(E) 12,000 + 40x

22. Which of the following must be greater than N if $0 < N < 1$?

(A) $\frac{1}{\sqrt{N}}$ (B) $\frac{\sqrt{N}}{2}$ (C) N^2

(D) 0.9N (E) $\frac{2N}{3}$

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23. A polygon with all sides equal is inscribed in a circle with center O. If R and S are adjacent vertices of the polygon, ∠ROS CANNOT equal

(A) 25°
(B) 24°
(C) 15°
(D) 9°
(E) 8°

24. If 125 per cent of m is equal to 80 per cent of n and $n \neq 0$, then $\frac{m}{n}$ =

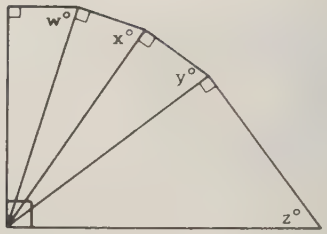
(A) $\frac{16}{25}$

(B) $\frac{4}{5}$

(C) 1

(D) $\frac{5}{4}$

(E) $\frac{25}{16}$



25. In the figure above, w + x + y + z =

(A) 180
(B) 210
(C) 240
(D) 270
(E) 360

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

35 QUESTIONS

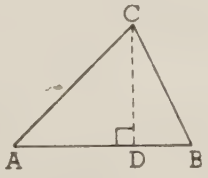
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Circle of radius r: Area = πr^2; Circumference = $2\pi r$
The number of degrees of arc in a circle is 360.
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Definitions of symbols:

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If ∠CDA is a right angle, then

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(2) $AC^2 = AD^2 + DC^2$

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---

1. If $3y - 2 = 13$ and $x + y = 4$, then $x =$

(A) -1 (B) $-\frac{1}{3}$ (C) $\frac{1}{3}$ (D) $\frac{23}{3}$ (E) 9

2. What is the average of $\frac{1}{8}, \frac{3}{8}, \frac{5}{8}$, and $\frac{7}{8}$?

(A) $\frac{3}{8}$ (B) $\frac{1}{2}$ (C) $\frac{5}{8}$ (D) 1 (E) 2

3. Which of the following figures can be drawn without lifting the pencil or retracing a segment?

I.



II.



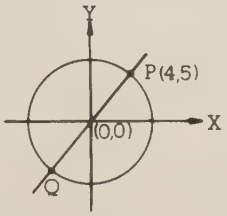
III.



(A) I only (B) II only (C) III only
(D) I and II (E) I and III

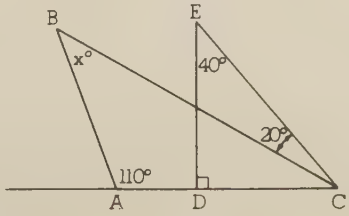
4. If p is a whole number such that the only positive divisors of p are 1 and p and if 20p is divisible by 6, then $p =$

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6



5. In the figure above, the coordinates of point P are (4, 5). If PQ is a diameter of a circle with center (0, 0), what are the coordinates of point Q ?

(A) (−5, −4) (B) (5, 4) (C) (−4, 5)
(D) (4, −5) (E) (−4, −5)



6. In triangles ABC and DEC above, $x =$

(A) 15 (B) 20 (C) 30 (D) 40 (E) 50

---

7. Which of the following is a pair of numbers that are NOT equal?

(A) $\frac{1}{8}, 0.125$ (B) $\frac{3}{7}, \frac{33}{77}$ (C) $2.2, \frac{11}{5}$

(D) $\frac{21}{63}, \frac{15}{45}$ (E) $\frac{44}{12}, \frac{28}{8}$

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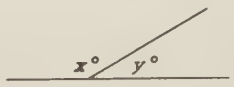
5

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

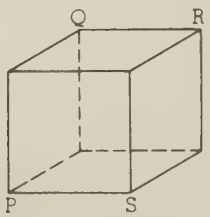
- A if the quantity in Column A is greater;
- B if the quantity in Column B is greater;
- C if the two quantities are equal;
- D if the relationship cannot be determined from the information given.

Notes:
1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. A symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	The number of hours in a day	The number of days in a month
9.	The value of $3x - 8$ when $x = 8$	The value of $3y - 8$ when $y = -8$



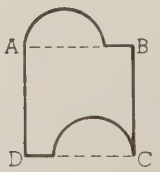
P, Q, R, and S are vertices of the cube.

	Column A	Column B
10.	Length of diagonal PR (not shown)	Length of diagonal QS (not shown)
11.	$0.001 + 8$	$0.008 + 1$

	Column A	Column B

Charge for laying tile is \$5 per square meter.

12.	Charge for tiling a rectangular region 32 meters by 20 meters	Charge for tiling a rectangular region 20 meters by 32 meters



ABCD is a square and the semicircles have equal diameters.

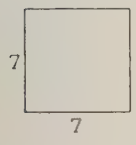
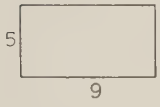
13.	Area of ABCD	Area of the figure outlined with the heavy solid line
14.	$9(7 + 3)$	$(3 \times 3)(2 + 8)$

$$\begin{cases} x + y = 5 \\ x - y = 5 \end{cases}$$

15.	y	5

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A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

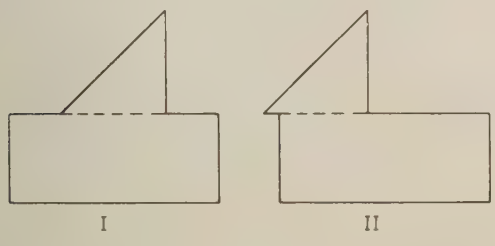
	Column A	Column B
16.	 Perimeter of the square	 Perimeter of the rectangle

p, q, and r are distinct positive integers.

17.	$p + q + r$	pqr
18.	The length of the hypotenuse of a right triangle with legs of lengths 4 and 2, respectively	5

$y < x < 0$

19.	x	$-2y$



A rectangle and a triangle are arranged without overlapping in the two ways shown.

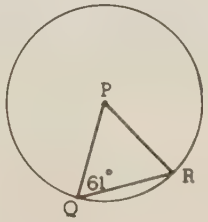
20.	Perimeter of arrangement I	Perimeter of arrangement II
21.	$1 - \frac{x}{y}$	$\frac{-1}{\frac{y}{y - x}}$

	Column A	Column B

Questions 22-23 refer to the following definition.

For all positive numbers n, $\boxed{n}$ represents the value of n^2 rounded to the nearest integer.

22.	$\boxed{2}$	The product of 4 and $\boxed{1}$
23.	$\boxed{n + 1}$	$\boxed{n} + 1$
24.	$\sqrt{0.049}$	0.07



Points Q and R are on the circle with center P.

25.	QR	PQ

$x \neq 1$

26.	$\frac{3x^2 - 3}{x - 1}$	$2x + 2$

The price of a hat increased by 25 per cent and then the new price was decreased by 20 per cent.

27.	The hat's original price	The hat's more recent price

GO ON TO THE NEXT PAGE

5

Solve each of the remaining problems in this section using any available space for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

28. If x is a number greater than 0, which of the following must be true?

(A) $x \div x = x$
(B) $x \div 1 = 1$
(C) $1 \div x = x$
(D) $1 \div 1 = x$
(E) $x \div x = 1$

29. The centers of two circles are 6 centimeters apart. If the circumference of one of the circles is 4π centimeters and the circumference of the other is 14π centimeters, which of the following is true?

(A) The circles do not intersect and no part of the small circle is within the large circle.
(B) The circles do not intersect and no part of the small circle is outside of the large circle.
(C) The circles intersect at one point only.
(D) The circles intersect at two points only.
(E) The number of points of intersection cannot be determined from the information given.

30. If $\frac{1}{3} + \frac{1}{4} + \frac{1}{5} = \frac{47}{N}$, then all of the following are integers EXCEPT

(A) $\frac{N}{24}$ (B) $\frac{N}{15}$ (C) $\frac{N}{12}$ (D) $\frac{N}{10}$ (E) $\frac{N}{6}$

31. If for all real numbers

$\langle a, b, c \times x, y, z \rangle = ax + by + cz$, then

$$\frac{\langle 2, 2, 5 \times 4, 1, 6 \rangle}{\langle -4, -1, -6 \times 2, 2, 5 \rangle} =$$

(A) −40
(B) −1
(C) 0
(D) 1
(E) 198

32. If $(2^5 - 2^4)(2^3 - 2^2) = 2^x$, what is x ?

(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

33. A grocer buys m crates of strawberries for q cents per crate. If each crate contains p liters which he sells at c cents per liter and if other expenses are disregarded, which of the following represents his total profit, in cents, on the strawberries?

(A) $\frac{pc}{m} - \frac{q}{m}$

(B) $pc - q$

(C) $\frac{pc}{m} - mq$

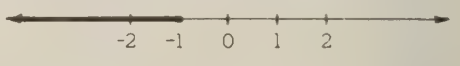
(D) $pc - mq$

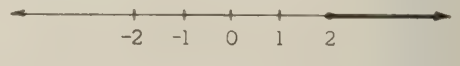
(E) $mpc - mq$

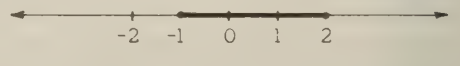
34. If $xy = -1$ and $(x + y)^2 = 10$, then $x^2 + y^2 =$

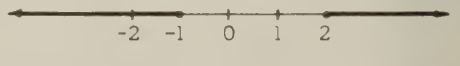
(A) 8
(B) 9
(C) 10
(D) 11
(E) 12

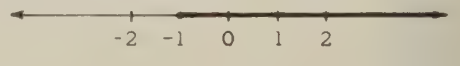
35. Which of the following is the graph of all x values such that $x \geq 2$ and $-x \leq -1$?

(A)


(B)


(C)


(D)


(E)


S T O P

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 6

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. VERTICAL: (A) even (B) horizontal (C) straight (D) open (E) parallel
2. SCURRY: (A) dawdle (B) calm (C) align (D) recline (E) capture
3. INNUMERABLE: (A) worthless (B) yielding (C) harsh (D) very few (E) moderately easy
4. IRRATIONAL: (A) curious (B) honest (C) pleasant (D) attentive to detail (E) guided by reason
5. STAGNANT: (A) deep (B) overt (C) crooked (D) flowing (E) tasteless
6. MOTIVATE: (A) deter (B) pester (C) misconstrue (D) alienate (E) minimize
7. DOUR: (A) unable to respond (B) cheerful in appearance (C) youthful (D) crafty (E) clean
8. HARDINESS: (A) vulnerability (B) superficiality (C) skillfulness (D) good fortune (E) poor judgment
9. ENHANCE: (A) vacate (B) insist (C) depreciate (D) volunteer (E) resist
10. AMASS: (A) shut down (B) give away (C) dress up (D) shine forth (E) show off
11. SQUALOR: (A) silence (B) elegance (C) rapidity (D) regularity (E) truthfulness
12. HACKNEYED: (A) liberated (B) fresh (C) afoot (D) logical (E) understood
13. RESCIND: (A) advise (B) falter (C) implement (D) demand (E) predict
14. PAUCITY: (A) ambition (B) eminence (C) intensity (D) hugeness (E) plenty
15. INCORPOREAL: (A) active (B) unusual (C) isolated (D) accurate (E) material

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. Common sense tells us that a person's ---- to use certain kinds of words ---- past experiences.

(A) desire. .recovers (B) refusal. .decides (C) wish. .determines (D) tendency. .reflects (E) inability. .inhibits

17. Only a ---- in the field will comprehend this highly ---- material.

(A) veteran. .general (B) critic. .complimentary (C) novice. .intricate (D) master. .conventional (E) specialist. .sophisticated

18. The harshness of political struggle ---- Ralph but at the same time fascinated him, so that he could neither ---- the battle nor fully commit himself to it.

(A) frightened. .dread (B) inspired. .comprehend (C) repelled. .abandon (D) intrigued. .forsake (E) annoyed. .influence

19. By attempting to ---- all the writer's inconsistencies in form, the translator distracts the attention of the general reader, who is not concerned with such close accuracy.

(A) improve (B) reproduce (C) equalize (D) interpret (E) omit

20. If the world's present known reserves of fossil fuels could be ----, construction of nuclear power plants would doubtless be less ----.

(A) augmented. .precipitate
(B) depleted. .expedient
(C) husbanded. .regulated
(D) recycled. .reliable
(E) discovered. .hazardous

GO ON TO THE NEXT PAGE

6

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

Americans disapproved a preliminary draft of the Declaration of Independence that indicted the king of England for waging "cruel war against human nature itself, violating its most sacred rights of life and liberty in the persons of a distant people who never offended him, captivating and carrying them into slavery in another hemisphere, or to incur miserable death in their transportation thither." The Americans were themselves not ready to abolish slavery and instead used the words "all men are created equal," which did not commit them to any specific action in regard to slavery. This statement represented a goal that some Americans hoped to achieve some day rather than a plan most of them were willing to act upon immediately.

Blacks were at first barred from serving in the revolutionary army although some blacks had already involved themselves in the struggle. Fearing that blacks would enlist in the British army, which welcomed them, and facing a manpower shortage, the Continental army soon began to accept free blacks. Runaway slaves, however, joined the British. More than 30,000 slaves ran away in 1778 alone, preferring military service to slavery. The states started to enroll both free and enslaved blacks, and finally Congress authorized military service for slaves, who were to be emancipated in return for their service.

After the war a few blacks achieved material success and some owned slaves, but the vast majority knew only poverty. Forbidden to settle in some areas and segregated in others, free blacks were targets of prejudice and discrimination. In the South, they were denied freedom of movement, restricted in their choice of occupation, and forbidden to associate with whites or slaves. Whites could legally challenge their freedom at any time, and free blacks lived in constant danger of being enslaved. In both the North and the South, they were denied political power and were regularly the victims of mobs.

Some Americans, Washington and Jefferson among them, advocated the gradual emancipation of slaves, and in the nineteenth century, a movement to abolish slavery grew in importance. A few white abolitionist leaders wanted full equality for blacks, but others sought only to eliminate the institution itself. And some antislavery societies, fearing that the inclusion of black members would unnecessarily offend those who were unsympathetic with abolitionist principles, denied entrance to black abolitionists.

Most Americans were, in fact, against abolishing slavery. They refused to rent their halls for antislavery meetings. They harassed abolitionist leaders who sought to educate white and black children together. They attacked those involved in the movement. Mobs sometimes killed abolitionists and destroyed their property.

21. The author implies that, if the Americans had indicted the king of England according to their original plan, their action would have been

(A) hypocritical (B) underhanded
(C) unjustified (D) futile (E) pragmatic

22. The author is primarily concerned with

(A) describing the status of blacks immediately before and after the Revolutionary War
(B) suggesting ways that blacks can help solve the problems of the United States
(C) showing the social and political achievements made by blacks since the founding of the United States
(D) tracing the effect that blacks have had on the development of the United States since the Revolutionary War
(E) showing that the abolitionists were ineffective in their efforts to improve the condition of blacks

23. According to the passage, blacks were denied entrance into antislavery societies because

(A) the black abolitionists were too radical for the whites
(B) such societies were against the law
(C) the abolitionists were afraid of public opinion
(D) most blacks were unwilling to associate with whites
(E) the societies were afraid that white members would misunderstand the presence of black members

24. It can be inferred that in the first half of the nineteenth century under United States law free blacks

(A) could not permanently reside in any one place
(B) had fewer rights than did slaves
(C) could be deported to another country without warning
(D) received as fair treatment as poor whites
(E) could be bound over into slavery

25. According to the author, the words in the Declaration of Independence, "all men are created equal," were meant to represent

(A) an objective (B) a pretext
(C) a provocation (D) an overstatement
(E) a rationalization

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Since the days of Copernicus, the greatest change that has occurred in our way of thinking about the universe is the modern conception of the universe as finite. This does not mean merely that there is a finite amount of matter in space, but that space itself is finite. The instant impulse that most lay people have to reject this statement as absurd is really an indication that they are unconsciously dominated by certain assumptions. These assumptions can be isolated and analyzed and shown to be logically arbitrary. The analysis is now about a hundred years old. It began with the invention of the first non-Euclidean geometry.

About a hundred fifty years ago Lobachevski, a Russian, and Bolyai, a Hungarian, found, independently of one another, that Euclid's geometry is not a logical necessity. This is not to say that Euclid's geometry is incorrect. If we start by accepting Euclid's axioms, then we must also accept the whole superstructure Euclid built on them. But the fact is that it is not necessary to accept Euclid's axioms. Quite different axioms can be postulated and a perfectly self-consistent system of geometry can be built on them. It then becomes a matter of experiment to determine which system of geometry best fits our experience.

The invention of the non-Euclidean geometries is one of the most remarkable feats in the intellectual history of humanity. For two thousand years Euclid's axioms had reigned unchallenged. That they were "necessary truths," true even for God, was admitted by all the philosophers. Merely to wonder whether these could be transcended was an effort of extraordinarily imaginative daring. And to translate this skepticism into the creation of a new, coherent, and complete system of geometry was a wonderful and exhilarating achievement of the free human mind.

Gradually the science of deductive geometry has acquired a new status. The question has arisen as to which geometry we are to apply to actual, physical space, the space in which all the millions of stars exist. The question is a matter of measurement. Our measuring instruments may be rods and clocks or rays of light and vibrating atoms. Which geometry best describes the observed behavior of the stars? A particular form of geometry invented by the German geometer Riemann best meets the facts of the case. Einstein was the first to realize this and he applied this geometry with immense success. Scientists are now convinced that we live in a space governed not by the laws of Euclid's geometry, but by the laws of Riemann's geometry.

26. According to the passage, Lobachevski and Bolyai discovered that

(A) there were errors in Euclid's axioms
(B) there is an infinite amount of matter in space
(C) Euclid's axioms could be applied to Copernicus' theories of the universe
(D) Riemann's geometry was applicable to physical space
(E) a consistent system of geometry other than Euclid's could be developed

27. According to the passage, Einstein was the first to recognize the

(A) use of vibrating atoms as measuring devices
(B) logical necessity of Euclid's axioms
(C) absurdity in the theory that the matter in space is finite
(D) possibility that Euclid's axioms could be revised
(E) importance of applying Riemann's geometry to actual, physical space

28. By "this skepticism" (line 35), the author means

(A) denying the existence of non-Euclidean geometries
(B) questioning whether Euclid's axioms were the only possible ones
(C) doubting people's ability to devise new systems of geometry
(D) resisting scientific pressure for the development of new philosophies
(E) criticizing the accuracy of the axioms in Euclid's geometry

29. The content of the passage primarily answers which of the following questions?

(A) How is deductive geometry similar to geometry based on Euclid's axioms?
(B) How did Lobachevski and Bolyai develop new systems of geometry?
(C) Why have people assumed that Euclid's geometry is a logical necessity?
(D) Why has our concept of space changed since the days of Copernicus?
(E) How do the laws of Riemann's geometry apply to the measurement of actual, physical space?

30. We can infer from the passage that Euclidean geometry is

(A) inconsistent
(B) unnecessarily complex
(C) no longer applicable
(D) not completely adequate for describing space
(E) more abstract than other geometries and therefore too restrictive for modern science

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6

Select the word or set of words that best completes each of the following sentences.

31. From a high-flying jet, the earth appears virtually ----; that is, rolling land is ironed out and the land's general ---- are almost eliminated.

(A) flat. .contours
(B) uninteresting. .colors
(C) round. .markings
(D) unrecognizable. .features
(E) motionless. .guidelines

32. To submit such ---- testimony is to ---- the defendant while pretending to prove his innocence.

(A) arbitrary. .acquit
(B) implausible. .incriminate
(C) airtight. .exonerate
(D) favorable. .sentence
(E) conflicting. .confuse

33. Well-intentioned ---- all too often defeats its basic purpose by destroying the recipient's self-reliance.

(A) benevolence (B) modesty (C) negligence (D) arrogance (E) gratitude

34. A man of great sensibility, the writer ---- the laying bare of his first work; he had put into the book everything he had that was good, true, and beautiful, and it was regarded as ----.

(A) thrilled to. .nonsense
(B) shrank from. .artistry
(C) suffered at. .drivel
(D) glossed over. .caricature
(E) agonized over. .abstraction

35. The union leader made the ---- statement that no union member, not even the maintenance crew, would be allowed to cross the picket line.

(A) discreet (B) ambiguous (C) fallacious (D) obsequious (E) categorical

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. SUBMARINE : SHARK :: (A) sled : dog
(B) plow : horse (C) caravan : camel
(D) airplane : hawk (E) subway : tiger

37. MAGNIFY : SIZE :: (A) pacify : blandness
(B) rectify : generosity (C) dignify : wisdom
(D) fortify : strength (E) verify : fear

38. FORUM : DISCUSSION :: (A) casino : gambling
(B) palette : painting (C) library : printing
(D) itinerary : traveling (E) armory : fighting

39. CRUNCH : SNOW :: (A) wade : beach
(B) ripple : sand (C) whistle : wind
(D) slosh : water (E) drizzle : rain

40. PROTEST : DISCONTENT :: (A) anger : indecision
(B) inaction : apathy (C) mockery : contrition
(D) sympathy : misery (E) supremacy : anarchy

41. BRAWN : WEIGHT LIFTER :: (A) fame : painter
(B) wound : doctor (C) uniform : soldier
(D) obedience : pioneer (E) tact : diplomat

42. BRINK : PRECIPICE :: (A) bank : river
(B) tree : forest (C) lake : ocean
(D) tunnel : mountain (E) whirlpool : waterfall

43. DAREDEVIL : RECKLESS :: (A) virtuoso : careless
(B) hostage : lawless (C) brute : helpless
(D) prodigal : thriftless (E) wizard : relentless

44. RACONTEUR : ANECDOTE :: (A) orator : speech
(B) lecturer : applause (C) actor : monologue
(D) operator : message (E) comedian : dialogue

45. APOCRYPHAL : AUTHENTICITY ::
(A) absurd : novelty (B) protean : variety
(C) unseemly : reality (D) illicit : legality
(E) ordained : sanction

IF YOU FINISH BEFORE TIME IS CALLED, YOU MAY CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 0X

VERBAL		MATHEMATICAL	
Section 1	Section 6	Section 3	Section 5
1. C	1. B	1. A	1. A
2. D	2. A	2. E	2. B
3. E	3. D	3. A	3. E
4. B	4. E	4. D	4. B
5. B	5. D	5. E	5. E
6. A	6. A	6. E	6. D
7. A	7. B	7. B	7. E
8. C	8. A	8. B	*8. B
9. E	9. C	9. A	*9. A
10. E	10. B	10. C	*10. C
11. A	11. B	11. C	*11. A
12. D	12. B	12. A	*12. C
13. A	13. C	13. D	*13. C
14. B	14. E	14. D	*14. C
15. A	15. E	15. D	*15. B
16. E	16. D	16. B	*16. C
17. C	17. E	17. A	*17. D
18. D	18. C	18. D	*18. B
19. B	19. B	19. B	*19. B
20. A	20. A	20. B	*20. B
21. C	21. A	21. C	*21. D
22. D	22. A	22. A	*22. C
23. D	23. C	23. A	*23. D
24. E	24. E	24. A	*24. A
25. D	25. A	25. D	*25. B
26. E	26. E		*26. D
27. D	27. E		*27. C
28. C	28. B		28. E
29. D	29. D		29. D
30. C	30. D		30. A
31. B	31. A		31. B
32. A	32. B		32. E
33. A	33. A		33. E
34. D	34. C		34. E
35. B	35. E		35. B
36. E	36. D		
37. B	37. D		
38. A	38. A		
39. E	39. D		
40. C	40. B		
	41. E		
	42. A		
	43. D		
	44. A		
	45. D		

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. The SAT-verbal test has 85 questions with five answer choices each. If, for example, a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty among different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing SAT scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the first two characters (number and letter) of the form code you marked in item 3 on your answer sheet are the same as the form code printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 6

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the next page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 6* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals. Enter the resulting figure on the worksheet.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on the back cover. Enter this figure on the worksheet. (The SAT-verbal score you just recorded and your reported SAT-verbal score should be identical. If not, see the paragraph at the bottom of the next page.)

SAT-Mathematical Sections 3 and 5

Step A: Count the number of correct answers and the number of incorrect answers for *section 3* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 5* and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals. Enter the resulting figure on the worksheet.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.) Enter the resulting figure on the worksheet.

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on the back cover. Enter this figure on the worksheet. (The SAT-mathematical score you just recorded and your reported SAT-mathematical score should be identical. If not, see the paragraph at the bottom of the next page.)

SAT SCORING WORKSHEET

FORM CODE 0X

SAT-Verbal Sections

A. Section 1: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 6: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. SAT-verbal reported scaled score (See the conversion table on the back cover.) [ ] SAT-verbal score

SAT-Mathematical Sections

A. Section 3: _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal A

B. Section 5: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____ no. incorrect) = _____ subtotal B

C. Section 5: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____ no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. SAT-mathematical reported scaled score (See the conversion table on the back cover.) [ ] SAT-math score

Should you have any questions on these scoring instructions, you may call the phone number below. If, after following the above scoring directions and checking your work at least twice, your results disagree with the SAT-verbal or SAT-mathematical score reported on your ATP Student Report, you may request rescoring of your answer sheet. If rescoring confirms that an error had been made (resulting in either higher or lower scores than those originally reported), corrected reports will be sent to all recipients of your original scores. Please send your request to:

College Board ATP
Box 592
Princeton, NJ 08541
Attention: Rescore Request
Telephone: (609) 883-8500

Please indicate whether it is your SAT-verbal or SAT-mathematical score, or both, that you want to be rescored. When you write, please include a copy of this scoring worksheet on which you did your calculations.

Score Conversion Table
Scholastic Aptitude Test
Form Code 0X

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	460	610
84	790		39	450	600
83	780		38	440	590
82	770		37	440	590
81	760		36	430	580
80	760		35	420	570
79	750		34	420	560
78	740		33	410	550
77	730		32	400	540
76	720		31	400	530
75	720		30	390	530
74	710		29	380	520
73	700		28	380	510
72	690		27	370	500
71	680		26	360	490
70	680		25	350	480
69	670		24	350	470
68	660		23	340	460
67	650		22	330	460
66	640		21	330	450
65	640		20	320	440
64	630		19	310	430
63	620		18	310	420
62	610		17	300	410
61	600		16	290	400
60	600	800	15	290	400
59	590	790	14	280	390
58	580	780	13	270	380
57	580	770	12	260	370
56	570	760	11	260	360
55	560	750	10	250	350
54	560	740	9	240	340
53	550	730	8	240	330
52	540	720	7	230	330
51	530	710	6	220	320
50	530	700	5	220	310
49	520	690	4	210	300
48	510	680	3	200	290
47	510	670	2	200	280
46	500	660	1	200	270
45	490	660	0	200	270
44	490	650	– 1	200	260
43	480	640	– 2	200	250
42	470	630	– 3	200	240
41	470	620	– 4	200	230
			– 5	200	220
			– 6	200	210
			– 7 or below	200	200